How Much Leverage Is Right?

Growth, Ruin, and Volatility Across Asset Classes

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Abstract. One identical levered machine is run over the longest free daily histories of twenty return series: constant leverage, daily rebalancing, financing at the Treasury bill rate plus a class spread, and liquidation the first time a close breaches a stated maintenance covenant. The growth-optimal leverage, the multiple that maximises long-run growth under a stationary block bootstrap, is a small multiple everywhere: 2.5 times for the US equity market, against a Gaussian Kelly fraction of 2.64 computed on the same moments; 3 times for gold; 1.75 times for the reconstructed 10-year Treasury; 1 to 1.75 times for the single-name panel; 1 to 1.25 times for the commodity rows; and 1.25 times for bitcoin. The simulated optimum is at or below the Gaussian fraction in fifteen of the twenty rows. The exceptions are the rows whose Kelly fraction is below 1, where the optimum sits at the grid floor of 1 time, and one liquidation artefact, the yen at its admissibility cap. The survival-optimal leverage, the most leverage that keeps the five-year probability of liquidation or ruin at or below 1 percent and the probability of a 50 percent drawdown at or below 5 percent, is lower, and for most rows it is below 1 time. At 1 time the drawdown condition alone binds: the unlevered five-year probability of a halving is 10 percent for Coca-Cola, 42 percent for silver, and 87 percent for bitcoin. The pre-committed unifying test fails for both optima: optimal leverage times volatility is not constant across classes (coefficient of variation 0.87 for the growth optimum and 1.09 for the survival optimum), so the cross-class answer is not a volatility rule alone; the tails, the drawdown tolerance, and the maintenance covenant each carry part of it. Every ruin probability reported is a lower bound, because margin is checked at daily closes and never intraday.

Keywords: leverage; Kelly criterion; margin; drawdown; ruin probability; volatility drag; stationary bootstrap

JEL codes: G11, G12, G32, C15, G01.

Non-technical summary. How much borrowed money can an investor hold against an asset before the borrowing itself becomes the risk? This paper answers with one identical machine for every asset: borrow at the Treasury bill rate plus a spread, hold a fixed multiple of the asset, rebalance daily, and accept a forced sale the first time a margin rule is breached. The machine runs over the longest free daily price histories available: US shares from 1926, government bonds from 1962, currencies, oil, metals, and bitcoin. Two numbers come out for each asset. The growth number is the multiple that compounds a stake fastest across a decade of resampled history. The survival number is the most borrowing that keeps two promises: at most a 1 percent chance of forced exit or wipeout within five years, and at most a 5 percent chance of watching the stake halve. The growth numbers are small: 2.5 times for US shares, 3 times for gold, between 1 and 1.75 times for individual company shares. The survival numbers are smaller, and for most assets they are below 1, meaning no borrowing at all: an unborrowed stake in a typical large company share already carries about a one-in-ten chance of halving within five years, and the promise, not the lender, is what fails. Multiplying the answers by each asset's volatility does not make them agree, so the differences between assets are not explained by how much they move alone. The shape of an asset's worst days, the drawdown a holder can honestly promise to sit through, and the margin rule itself each move the answer. Eight well-known blow-ups,

from the 1929 margin collapse to the 2022 nickel squeeze, are checked against the same machine; each is tied to the one quantity it isolates, among them the one-day ruin bound, the cost of financing terms that worsen inside the event, and the difference between a solvency problem and a collateral problem.

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1 Introduction

How much leverage is right? The question is usually answered with a rule of thumb (never more than 2 times; futures traders use 10; Kelly says a number most people refuse) or with an anecdote (1929, LTCM, Archegos). This paper answers it the same way for every asset class at once: one identical levered machine, run over the longest daily histories the public record provides for free, with the financing rate, the maintenance covenant, and the exit rules stated in advance and held constant across classes. The machine is deliberately simple. An investor holds a constant leverage multiple of a single asset, rebalanced daily, financed at the Treasury bill rate plus a class spread, and is liquidated by the broker the first time a close breaches the maintenance requirement. Against that machine the paper computes two optima per class. The growth-optimal leverage maximises the long-run growth rate of the stake. The survival-optimal leverage is the most leverage at which two promises can still be kept with pre-stated confidence: the probability of forced exit or ruin within five years stays at or below 1 percent, and the probability of watching the stake halve stays at or below 5 percent.

Three findings survive every robustness variant the paper runs. First, the growth-optimal leverage is below the Gaussian Kelly fraction computed on the same data for nearly every class, and the gap is priced by the tails: the formula sees variance, the resampled paths see the worst day in the sample, and at high multiples one such day decides the decade (for the equity market, Kelly on the sample's moments is 2.64 times; the bootstrap optimum is 2.5 times). Second, the survival optimum is a much lower number than the growth optimum, and for most assets in the study it does not exist at any leverage on the grid, including 1 times: an unlevered stake in a 23-percent-volatility single name already carries a 10 percent probability of halving within five years, so the binding constraint for such assets is the drawdown promise, not the lender. Third, the pre-committed unifying test (RQ4) fails for both optima: leverage times volatility is not constant across classes (growth: coefficient of variation 0.87, max/min 15.4; survival: coefficient of variation 1.09, with the product at zero in seven of twelve core classes because the survival optimum itself is below 1 time), so the honest summary is that volatility orders the cross-class pattern without completing it, and what remains is the shape of the tails, the drawdown tolerance, and the maintenance covenant, which is asset-class information a

volatility-only rule throws away.

The eight case studies are part of the measurement, not decoration. Each is tied to a quantity the machine prices: 1929 to the survival leverage of a slow crash at era margin terms (2.0 times, surviving with 98 percent of the stake gone); 1987 to the one-day ruin bound; LTCM to the error of sizing leverage against the wrong volatility; 2008 to the cost of financing terms that worsen inside the event (survival leverage on the crisis equity window falls from 3.0 to 2.25 times when the spread and maintenance step up after a 10 percent drawdown); Archegos to the survival leverage of concentrated single names; the 2022 LDI episode to the difference between a solvency constraint and a collateral constraint; the 2022 nickel squeeze to the short side's ruin bound; and Volmageddon to the boundary of the method, a convex underlying whose distribution is manufactured by the leveraged products that trade it.

The paper is a retrospective simulation study. Its design, tolerances, and verdict rules were fixed before the simulations were run and are recorded in the build notes; no part of it is a pre-registered plan, and its ruin probabilities are lower bounds for a reason stated wherever they are used (margin is checked at closes, and the real world also checks intraday).

The rest of the paper proceeds as follows. Section 2 describes how leverage is actually supplied under the four contracts the machine abstracts (margin lending, futures margin, swaps, and leveraged funds, with repo and LDI as the financing-spiral cases). Section 3 places the design in the growth-optimal, leverage-cycle, and drawdown literatures and states the simulation method. Section 4 documents the data, including two source substitutions and one demotion (corporate bonds) that the free record forced. Section 5 specifies the machine and the four research questions as computations. Section 6 reports the results and the RQ4 verdicts. Section 7 works the eight cases on a standardised basis. Section 8 states limitations, the falsifiability conditions, and the robustness summary. Section 9 concludes. Appendix A is the data appendix; Appendix B records the implementation checks and the regime detail.

2 How leverage is actually supplied

The machine in this paper is an abstraction of four real contracts. The differences between them decide which of the paper's two engines, the liquidation rule or the no-liquidation variant, is the right model of a given exposure, so the contracts are worth stating plainly.

2.1 Margin lending against securities

A broker lends a customer part of the purchase price of a security and takes the security as collateral. In the United States, Regulation T has set the initial margin for equities at 50 percent since 1974, so a margin account can open at most 2 times leverage. Exchange and FINRA maintenance rules then require equity of at least 25 percent of the position's current value; many brokers impose higher house requirements. The maintenance rule is a covenant, not a price: the moment equity falls below the required fraction at a valuation point, the broker may liquidate without notice. Two consequences follow for the arithmetic of this paper. First, a margined equity position cannot be held above 4 times leverage at all, because equity is 25 percent of position value at 4 times and any downtick breaches the covenant; the position is born in violation. Second, the covenant binds through the path, not at the horizon: what matters is the worst close, not the final price. The 25 percent maintenance level is the base case for equity-type exposures in this paper, and the admissibility cap it implies (leverage at most $1/m$) is treated as part of the contract, not as a modelling choice.

Before 1934 there was no federal initial margin rule. Brokers commonly lent 90 percent of purchase price against stock collateral, funded in turn by the call money market, where banks and corporations lent at rates that could spike violently when credit tightened (Observed, B). The 1929 case in Section 7 runs on those era terms: 10 percent margin, a maintenance practice far looser than the modern rule, and financing whose price moved against borrowers in the event.

2.2 Futures margin

An exchange-traded futures contract is not a loan. The clearing house requires initial margin, a performance bond set as a fraction of contract value (for equity index and energy futures, typically in the range of 3 to 12 percent), and settles gains and losses in cash every day through variation margin. Leverage of 10 times or more is therefore ordinary and admissible, which is why this paper's futures-style exposures (energy, metals, agriculture) carry a 10 percent maintenance parameter and a cap of 10 times. The binding constraint is liquidity, not solvency arithmetic: variation margin must be posted in cash on the day, and a position that cannot fund its daily settlement is closed by the clearing broker regardless of its holder's view of value. The nickel case in Section 7 is this constraint failing at scale, and being suspended by the exchange rather than settled.

2.3 Swaps and synthetic leverage

A total return swap gives its holder the return on a stock or index in exchange for a financing leg, with no purchase and, historically, no consolidated disclosure of the holder's aggregate position. The dealer's protection is a margin agreement negotiated bilaterally: initial margin posted by the holder, variation margin as the position moves. The Archegos case in Section 7 turns on the level of that negotiated margin: one prime broker had cut its requirement from the 15 to 25 percent range to 7.5 percent, which multiplied the leverage the same capital could support, and no counterparty could see the aggregate exposure across six dealers (Observed, B). In the language of this paper, synthetic leverage replaces the maintenance parameter m with a negotiated number that competition can push down, and the liquidation rule fires at several brokers on the same day.

2.4 Leveraged funds and the no-liquidation contract

A leveraged exchange-traded fund or note holds the leverage inside the product. The investor's loss is capped at the stake; there is no margin call on the investor; the fund rebalances daily to its target multiple and bears the financing and the volatility drag itself. This is the contract this paper's no-liquidation variant models: the equity path compounds at constant leverage until it does not, and ruin can arrive only through the path (a single day worse than $-1/L$, or the slow bleed of drag against a sideways market), never through a covenant. The Volmageddon case in Section 7 is the boundary of this contract: an inverse product is a leveraged position in a convex underlying, and its termination clause, not any margin rule, ended it in a day.

2.5 Repo, haircuts, and liability-driven investment

Secured financing through repurchase agreements prices leverage as a haircut: the lender advances $(1 - h)$ against collateral of value 1, so the borrower runs leverage of $1/h$ on that collateral. Gorton and Metrick (2012) document that average haircuts on structured debt were near zero in early 2007 and rose to roughly 27 percent on average across their collateral categories in 2008, with the weakest

categories near 45 percent; a book financed at a zero haircut that must be refinanced at 27 percent has to shrink by more than a quarter or find the cash. Section 7 models this as a financing spiral: after a 10 percent drawdown, the spread and the maintenance requirement step up together, which is the haircut mechanism translated into the machine's two financing parameters.

Liability-driven investment (LDI) inverts the usual problem. A pension scheme holding long-dated liabilities hedges them with long gilts and interest rate swaps obtained through repo and derivatives, posting collateral against the hedge rather than owning the bonds outright. The hedge can be exactly right for solvency (liability values fall as yields rise, matching the hedge loss) and still fail on liquidity, because the collateral call is in cash, in days, while the offsetting gain is a smaller future pension bill. The September 2022 gilt episode is the cleanest modern observation of the difference between being hedged and being funded, and Section 7 treats it in those terms.

3 Related literature

Four literatures meet in this paper, and each supplies one piece of the design.

Growth-optimal portfolios. Kelly (1956) showed that maximising the expected logarithm of wealth maximises the long-run growth rate and, with probability one, eventually outperforms any other fixed strategy; Breiman (1961) gave the result its rigorous form, and Thorp (2006) documents its practical career. The same literature is unusually honest about the costs: full Kelly leverage produces drawdowns that most holders will not tolerate, and MacLean et al. (2011) collect the case for fractional Kelly rules on exactly that ground. The Gaussian benchmark used throughout this paper, $L = (\mu - r)/\sigma^2$ on the sample's own moments, is the continuous-time Kelly fraction for a single risky asset; everything interesting in the results is a deviation of the simulated optimum from that line, in a direction the return distribution's tails explain. Grossman and Zhou (1993) add the constraint this paper's survival criterion resembles: maximising growth subject to a floor on wealth changes the optimal policy and generally lowers leverage.

Leverage cycles and margin spirals. Geanakoplos (2010) formalises leverage as an equilibrium object set by margin requirements, so that bad news and tighter margins reinforce each other. Brunnermeier and Pedersen (2009) split the mechanism into a loss spiral and a margin spiral, both of which appear in this paper's 2008 variant, where the financing terms themselves worsen after a drawdown. Adrian and Shin (2010) document that intermediary leverage is procyclical in the data: balance sheets expand when measured risk is low. The design of this paper holds the investor's leverage constant by rule, which is the opposite pole from the Adrian and Shin balance sheet; the comparison is deliberate, because it separates what leverage does to a holder who does not adjust it from what adjusting it does.

Drawdowns and survival as the stated objective. A distinct line of work treats the maximum drawdown, not variance, as the risk to be controlled, from the drawdown-constrained growth problem of Grossman and Zhou (1993) to drawdown-based performance and risk measures used in the managed-futures industry. This paper's survival-optimal leverage belongs to that line: it is the highest leverage at which two pre-stated bad events (forced exit or ruin, and a halving of the stake) stay below pre-stated probabilities. Nothing in the criterion references utility, which is the point: it is a standard a holder can state to a lender, a trustee, or themselves in advance.

Simulation method. With daily histories of up to a century, the sampling distribution of path statistics is obtained two ways: directly, from every overlapping historical window, and by resampling. The resampling uses the stationary bootstrap of Politis and Romano (1994), which draws blocks of random length (geometric, mean 21 trading days in the base case) so that the resampled series is

stationary and short-horizon dependence, volatility clustering included, survives in the paths. Block lengths of 5 and 63 days are run as sensitivity. The two engines are never averaged; where they disagree, the paper reports both and says which it believes and why.

4 Data

All series are free and public; no paid data is used anywhere in the paper. Table 1 summarises each return series; Appendix Table 5 gives the producer, the series identifier or construction, the observation count, and the sample for every series, and the data manifest (series, source, served start and end, pull date) is preserved with the replication files. The pull date for all series is 26 October 2026 (OBSERVED, A).

Equities. The US equity market is the value-weighted total return series from the Kenneth R. French Data Library (daily, from July 1926), the longest free daily equity history and the paper's workhorse. The Nasdaq Composite (FRED: NASDAQCOM, from February 1971) is a price index; it carries no dividends, which biases its measured growth and its Kelly fraction downward relative to a total return series, and the bias is stated wherever the series is used. The five single names (Coca-Cola, Exxon Mobil, JPMorgan, Apple, Tesla) are adjusted closes, so their returns are total returns; samples begin at each name's listing or at the start of the served history (1970 for Coca-Cola and Exxon Mobil, 1980 for JPMorgan and Apple, 2010 for Tesla). The panel is five large, long-lived names chosen for history length, which makes it a survivor panel: every name in it is alive and large in 2026, and its risk figures are therefore lower bounds on the single-name class (Section 8). The design's fallback for the high-volatility equity leg, the French market-plus-SMB combination, is retained as a robustness series.

Two substitutions against the original plan are recorded here because the data appendix is where they belong. First, the plan's first-choice vendor for single names and futures (Stooq) verified this client and then refused the downloads; the series were taken instead from the Yahoo Finance chart API, adjusted close for equities, which is the better series for the purpose (the first choice serves unadjusted closes). Second, investment-grade corporate bonds: the ICE BofA US Corporate Index total return value as served by FRED begins only in October 2023, no free long daily corporate total return index was found in the ElkassabgiData catalogue either, and the design's resolution order therefore demotes corporates to a post-2023 exhibit (783 trading days). The bond class in the headline results is carried by the Treasury reconstruction.

Bonds. The 10-year Treasury series is a reconstruction (RECONSTRUCTED, A): a constant-maturity 10-year par bond repriced daily from the FRED DGS10 yield, with duration and convexity computed numerically from the yield itself and coupon accrual at the prior day's yield. Reconstructions invite distrust, so the series was validated before use against published 10-year Treasury total returns: it gives +9.3 percent for 2019, +10.4 for 2020, −4.1 for 2021, −16.4 for 2022, and +3.8 for 2023, within about a percentage point of the published figures in each year, the 2022 figure (the year the class exists for) matching the published −16 to −17 percent range.

Commodities. Energy prices are FRED spot series (WTI from 1986, Henry Hub natural gas from 1997). Metals and agriculture are front-month futures (gold, silver, copper; corn, wheat, soybeans, from August 2000), unadjusted for roll yield; the agriculture row is an equal-weight basket of the three grains' daily returns. Spot gold's free daily history ends in 2013 on the usual source and the design's resolution order assigns the futures series; pre-2000 daily gold is a stated gap.

Currencies. The currency rows are unhedged foreign cash from a US investor's perspective: the spot change in the FRED daily exchange rate plus the foreign 3-month deposit rate (OECD series via

Table 1: Summary statistics by return series. Mean and volatility are annualised from daily returns (bitcoin: calendar-day steps, 365). Worst 21d is the worst rolling 21-step cumulative return. All values Reconstructed from the sources in Appendix Table 5.

Series	Start	Obs.	Mean (% per year)	Vol.	Skew	Ex. kurt.	Worst day (%)
US equity market (French)	1926	26,317	10.9	17.1	-0.17	16.12	-17.4
Nasdaq Composite	1971	14,047	12.0	20.2	-0.11	9.44	-12.3
US Treasury 10y (recon.)	1962	16,187	5.9	7.3	0.22	5.32	-3.6
Crude oil (WTI)	1986	9,525	14.4	45.8	0.33	38.85	-51.3
Natural gas	1997	5,711	70.8	140.6	14.46	454.64	-75.4
Gold (futures)	2000	6,563	11.9	17.9	-0.30	5.87	-10.8
Silver	2000	6,564	14.8	33.1	-0.93	12.89	-31.3
Copper	2000	6,567	11.4	27.2	-0.31	8.32	-22.3
Agriculture basket	2000	6,515	7.9	23.2	0.08	2.93	-10.5
EUR/USD	1999	6,973	1.9	9.1	0.16	2.63	-2.9
GBP/USD	1971	13,994	5.8	9.4	-0.26	6.57	-7.8
USD/JPY	2002	6,159	0.1	9.8	0.46	4.72	-3.3
Bitcoin	2014	4,312	71.0	71.0	0.87	44.88	-53.8
Coca-Cola	1970	14,325	13.8	23.2	-0.07	13.32	-24.7
Exxon Mobil	1970	14,325	15.3	23.8	-0.01	12.38	-23.4
JPMorgan	1980	11,747	18.5	34.9	0.44	14.47	-27.7
Apple	1980	11,559	27.5	43.6	-0.36	18.77	-51.9
Tesla	2010	4,106	50.1	57.2	0.36	4.92	-21.1
US small-cap leg (Mkt+SMB)	1926	26,317	11.8	18.4	-0.28	14.61	-12.3
US IG corporates (exhibit)	2023	783	5.9	5.0	0.01	1.54	-1.4

FRED) accrued daily, against the paper's standard financing leg. The euro sample starts with the currency in 1999; sterling reaches 1971; the yen sample starts in April 2002, the served start of the Japanese deposit-rate series.

Crypto. Bitcoin is the FRED Coinbase reference rate (CBBTCUSD, from December 2014). It trades every calendar day, so its steps are calendar days and its annualisation uses 365; it is reported in the tables for completeness and quarantined from every cross-asset claim, per the design: an eleven-year, single-regime, survivor asset cannot bear weight in a comparison against a century of equities.

Financing. The financing leg is the 3-month Treasury bill rate (FRED: DTB3) as served, which begins 4 January 1954, not in 1934 as the design assumed; the served start governs, and the French daily risk-free rate (annualised) fills the pre-1954 financing for the one series (the French market) that reaches back that far. The design's per-class spreads (50 and 100 basis points) are added on top as specified in Section 5.

5 Methodology

5.1 The levered machine

Every asset class is run through one identical machine. An investor with equity $E_0 = 1$ holds a position of L times equity in a single asset, rebalanced back to constant leverage L at every close. The equity update on day t is

$$E_t = E_{t-1}\Big(1 + L\, r_t - (L-1)\, d_t\Big),$$

where r_t is the asset's total return on day t and d_t is the daily financing rate, the 3-month Treasury bill rate on that day plus a class spread (50 basis points per year for index, bond, currency, and commodity exposures; 100 basis points for single names), divided by 252. At $L = 1$ the financing term vanishes and the machine reduces to the asset itself, which is the first of several arithmetic checks the implementation was required to pass before any result was read (Appendix B lists the checks).

The liquidation rule is the maintenance covenant of Section 2, checked at each close after that day's move and financing, before rebalancing. If

$$\frac{1 + Lr_t - (L-1)d_t}{L(1 + r_t)} \;<\; m,$$

the position is liquidated at that close: the path ends, residual equity is frozen at its post-move value (floored at zero; it cannot be negative under a close-based check), and the event is recorded as a margin-call liquidation. The parameter m is 0.25 for equity, bond, currency, and crypto exposures and 0.10 for the futures-style exposures (energy, metals, agriculture). A leverage level with $1/L < m$ breaches the covenant at the first close almost surely; such levels are inadmissible under the rule, are flagged as such in the results, and are excluded from the optima of the liquidation engine. Their outcomes are reported from the no-liquidation variant instead, which is the contract (a leveraged fund) under which such exposure is in fact held. Ruin is defined independently of liquidation: equity at or below 1 percent of its starting value at any close. The grid is $L \in \{1.00, 1.25, \ldots, 3.00, 3.50, 4.00, \ldots, 10.00\}$, 23 points.

5.2 Two engines, never averaged

Raw windows. For each horizon (1, 5, and 10 years; 252, 1260, and 2520 trading days), every overlapping window of the asset's own history is run through the machine, and the full-sample single path is run as well. Raw windows preserve the actual sequence of history, including its particular ordering of crashes and recoveries; they are thin exactly where it matters (a century holds fewer than ten independent 10-year windows, and a 26-year futures history holds two).

Stationary bootstrap. Resampled paths of 2520 days are drawn from the asset's own (return, financing) day pairs by the stationary bootstrap of Politis and Romano (1994) with mean block length 21 trading days; 10,000 paths per configuration. Drawing day pairs rather than returns alone keeps the financing rate that actually prevailed on the drawn day attached to it, so high-rate eras carry their own financing cost into the resampled paths. Volatility clustering survives in blocks; the specific century-scale sequence of regimes does not, which is the honest difference between the engines and the reason both are reported.

5.3 The questions, as computations

RQ1, growth. The growth-optimal leverage is the grid point maximising the mean annualised log growth of equity at the 10-year horizon (bootstrap engine; the raw full-sample path gives the second engine's value). The benchmark is the Gaussian Kelly fraction $(\bar{r} - \bar{f})/s^2$ computed on the same sample's daily excess returns, with $\bar{f}$ the mean daily bill rate without spread.

RQ2, survival. The survival-optimal leverage at a horizon is the highest grid leverage at which two pre-stated tolerances both hold: the probability of margin-call liquidation or ruin is at most 1 percent, and the probability of a maximum drawdown of 50 percent or more within the horizon is at most 5 percent. The drawdown probabilities at 25 and 90 percent are tabulated alongside. Where no grid leverage satisfies the tolerances, including 1 times, the optimum is reported as below 1: the drawdown

tolerance is then the binding one, and the tables identify which condition binds. Probabilities of ruin at 2, 3, 5, and 10 times at the 5-year horizon are headline outputs, taken from the liquidation engine where the level is admissible and from the no-liquidation engine, marked, where it is not.

RQ3, financing and regime dependence. The base engines are re-run at financing spreads of 0 and 300 basis points (and, for the figure, at 150 and 500 basis points on 5,000 paths for the core assets). Named subsamples (pre-2000, 2000 to 2007, 2008 to 2009, 2010 to 2019, 2020, 2021 to present) and terciles of trailing 63-day realised volatility are bootstrapped separately for the core assets at 5,000 paths, and the two optima are recomputed within each.

RQ4, the unifying test. If one number governs all classes, it is not leverage but leverage times volatility: a position's risk scales with $L\sigma$. The test pre-commits its verdict rule. Across the core classes (excluding crypto, which is quarantined, and single names, which are idiosyncratic), compute $L^*\sigma$ for the growth optimum and for the 5-year survival optimum, with σ the asset's full-sample annualised volatility. The volatility answer *holds* for an optimum if the coefficient of variation of $L^*\sigma$ across classes is at most 0.30 and the ratio of largest to smallest is under 2. It fails otherwise. The rule was fixed before the simulation was run, and the verdict is reported for each optimum separately, wherever it lands.

5.4 What the machine leaves out

The machine rebalances daily at closes with no transaction cost beyond the financing spread, checks margin only at closes, and treats financing as a rate rather than a relationship that can be withdrawn. Each omission leans the same way: intraday margin checks would liquidate some paths the close-based rule saves (the October 1987 intraday path being the standard example), so every ruin and liquidation probability reported here is a lower bound on its real-world counterpart. Section 8 states the full limitation set, including single-name survivorship and the quarantine on crypto.

6 Results

Table 2 is the paper. Everything in this section is a reading of it and of Figures 1 to 6; the numbers are Modelled (bootstrap engine, base case) unless a second engine is named, grade B, with the raw-window values in parentheses in the table where the design calls for them.

6.1 Growth-optimal leverage is usually below the Gaussian benchmark

For the US equity market over the full sample, the Gaussian Kelly fraction on the sample's own moments is 2.64 times. The bootstrap growth optimum is 2.5 times (raw full-sample path: 2.5 times), and the shape of the curve (Figure 1) shows why the simulated optimum sits below the formula: the formula prices volatility as if returns were Gaussian, while the resampled paths contain the sample's actual worst days, and at leverage above the peak a single such day costs more growth than a year of drag saves. The same pattern, a simulated optimum at or below the Kelly fraction computed on the same moments, holds in fifteen of the twenty rows of Table 2. The exceptions are instructive: where the Kelly fraction is itself below 1 (WTI, natural gas, EUR/USD), the optimum sits at the grid floor of 1 time; Tesla's two values agree to rounding; and the yen's optimum is a liquidation artefact, treated separately below. The Treasury reconstruction is the extreme in the other direction: with volatility of 7.3 percent and thin tails by equity standards, its Kelly fraction is 2.72 times against a bootstrap growth optimum of 1.75 times, and its survival optimum of 2.25 times is the highest of the

headline rows. For a bond holder the binding constraint is not growth but the maintenance covenant and the drawdown tolerance below.

Two boundary results deserve stating plainly. First, for several high-volatility series the growth curve peaks at or near 1 times: leverage adds drag faster than it adds drift from the first step. Apple is the panel's clean example (growth optimum 1.0 times against a Kelly fraction of 1.24, because its history contains a −51.9 percent day, which at 2 times is a ruin event inside most resampled paths). Second, bitcoin's growth optimum of 1.25 times, on a sample with annualised volatility near 71 percent, is what the drag arithmetic predicts; it is reported for completeness and excluded from the cross-asset claims (Section 4).

One row must be read with its mechanism attached. The yen's bootstrap growth optimum prints at the 4 times admissibility cap, against a Gaussian fraction of −1.71: the yen sample's mean excess return is negative, and every leverage at which the position can actually be held loses growth. At the cap itself, a position is liquidated on its first downtick and the stake then sits in cash, so measured growth is about zero, which beats every holdable leverage without being one. The no-liquidation variant, which cannot exit, declines monotonically in leverage and peaks at the 1 time floor (Table 4). The same edge prints in the raw column for natural gas, copper, and bitcoin, whose full-sample raw optima sit at their caps for the same reason. Cap values produced by first-downtick liquidation are accounting values of the exit rule, not holdable optima, and no cross-asset claim in this paper rests on them.

6.2 Survival-optimal leverage is a different, lower number, and for some assets it is below one

The survival criterion (liquidation or ruin at most 1 percent, a 50 percent drawdown at most 5 percent, five-year horizon) binds much harder than growth. For the equity market the survival optimum is 1 times against a growth optimum of 2.5 times. For the Treasury reconstruction it is 2.25 times, the highest of the headline rows. For WTI crude it is below 1 times. The single-name panel median and the currency range (growth optima of 1.5 times for the panel median name, 1 to 4 times across the three currency rows) sit in the same low region, but the survival column sits lower than a small multiple: at the five-year horizon the survival optimum is below 1 time for thirteen of the twenty rows, including the Nasdaq, every one of the five single names, WTI, natural gas, silver, copper, the agriculture basket, bitcoin, and the small-cap leg. Only the equity market index, the Treasury reconstruction, gold, the three currency rows, and the corporate exhibit admit a survival optimum at or above 1 time. That result is not a modelling defect; it is the drawdown condition speaking. At 1 time the liquidation and ruin probabilities are effectively zero in those rows, while the probability of a 50 percent drawdown within five years is 10 percent for Coca-Cola, 11 percent for the Nasdaq, 42 percent for silver, 75 percent for WTI, and 87 percent for bitcoin. The tolerance, not the leverage, is what such assets cannot meet, and the tables identify the binding condition in each case.

The ruin columns of Table 2 are the practical core for a holder choosing a multiple. At five years, the equity market's ruin probability is 0.0 percent at 2 times and 0.0 percent at 3 times under the liquidation rule; above the 4 times admissibility cap the exposure cannot be held on margin at all, and the no-liquidation (leveraged-fund) contract gives 5.5 percent at 5 times and 58.6 percent at 10 times. Figure 2 shows the joint liquidation-or-ruin probability rising with leverage at the three horizons; Figure 3 shows the drawdown distribution that does the damage, and Figure 6 the median time under water, which lengthens quickly past the growth optimum: leverage beyond the peak is slower, and slower while submerged.

6.3 RQ4: is there only a volatility answer?

The pre-committed test multiplies each class's optimum by its volatility and asks whether the product is constant (verdict rule in Section 5.3). For the growth optimum, $L^*\sigma$ has a coefficient of variation of 0.87 and a max/min ratio of 15.4 across the core classes: the volatility answer fails for growth. For the 5-year survival optimum, the coefficient of variation is 1.09 and the max/min ratio is not finite, because the product is zero in seven of the twelve core classes, every class whose survival optimum is below 1 time: the volatility answer fails for survival as well. The growth failure is driven from both ends of the volatility range: natural gas, at 141 percent volatility with its growth optimum at the 1 time floor, has the largest product at 1.41, while EUR/USD and GBP/USD, whose optima also sit at the floor, have the smallest, at about 0.09. Figure 4 plots both products asset by asset. The reading the paper puts on the pair of verdicts is in the conclusion; the numbers themselves are stated here without decoration, as the falsification discipline requires.

6.4 Financing and regimes (RQ3)

The financing spread is a direct tax on growth and a slower tax on survival. Figure 5 traces the growth optimum from 0 to 500 basis points for the core assets: it falls as the spread widens wherever it sits above the 1 time floor, most steeply for the low-volatility ones (a 300 basis point spread is a minor cost against equity volatility and a major one against a Treasury's: the Treasury growth optimum falls from 2.75 times at a zero spread to 1 time at 300 basis points), while the survival optimum moves less, because survival is priced by the tails and the drawdown condition more than by the drift. Within regimes (Appendix Table 6), the equity market's two optima split in the 2008 to 2009 subsample: the growth optimum is 4 times against 2.5 times on the full sample, because the resampled crisis pool contains the 2009 rebound days as well as the crash, while the survival optimum is below 1 time, against 1 times on the full sample, 1.75 times in the 2010 to 2019 subsample, and 2.25 times in the low-volatility tercile. Leverage sized on the calm subsamples therefore fails the crisis pool on the survival criterion, which is the LTCM warning in this paper's arithmetic, and the stability falsifier of Section 8 is not triggered on the survival side. The high-volatility tercile lowers both optima for nearly every core asset. The 2020 subsample disciplines the growth point: a crash and a policy-driven recovery inside one pool of days gives an equity growth optimum of 2.25 times, close to the full sample's, because resampling that year draws its recoveries as often as its crash; its survival optimum is below 1 time as well.

7 Case studies

The eight cases are not illustrations. Each is tied to one quantity in the framework, and Table 3 reports the set on a standardised basis: effective leverage, the volatility the holder faced before the event, the adverse move realised, the time to distress, what happened to the financing terms during the event, and the highest leverage at which this paper's machine survives the same asset over the same window (Modelled, B, unless noted).

7.1 1929: the margin system itself

The quantity is the survival leverage of the 1929 to 1932 path at era terms. Margin practice before federal regulation let a buyer put down typically 10 percent and borrow the rest against the shares (Federal Reserve History, 2013) (Observed, B); brokers' loans reached about USD 8.5 billion

Table 2: Main results by asset. Growth-optimal leverage: stationary bootstrap, 10,000 paths, 10-year horizon (raw full-sample value in the adjacent column). Survival-optimal: highest leverage with P(liquidation or ruin) $\leq 1\%$ and P(drawdown $\geq 50\%$) $\leq 5\%$ at five years. Ruin probabilities at 5 and 10 times for assets whose maintenance cap (4 times at $m = 0.25$, 10 times at $m = 0.10$) is below those levels are from the no-liquidation variant, the leveraged-fund contract under which such exposure is held; all other ruin probabilities are from the liquidation engine. Modelled, B.

Asset	Vol. (%/yr)	Kelly	Growth-opt. (boot.)	(raw)	Survival-opt. (5y)	P(ruin in 5y, %) at 2x	3x	5x	10x	P(DD50) at surv.
US equity market (French)	17.1	2.64	2.5	2.5	1	0.0	0.0	5.5	58.6	4.8
Nasdaq Composite	20.2	1.87	1.75	1.75	<1	0.0	0.2	13.0	77.1	11.1
US Treasury 10y (recon.)	7.3	2.72	1.75	1.75	2.25	0.0	0.0	0.0	2.4	4.9
Crude oil (WTI)	45.8	0.54	1	1	<1	16.1	46.7	39.3	1.3	74.9
Natural gas	140.6	0.35	1	10.0	<1	71.9	71.8	31.4	6.1	99.9
Gold (futures)	17.9	3.14	3	3	1.25	0.0	0.0	1.0	0.1	4.8
Silver	33.1	1.18	1	1	<1	0.0	5.2	26.1	0.4	41.9
Copper	27.2	1.29	1.25	10.0	<1	0.1	2.9	42.5	0.1	27.4
Agriculture basket	23.2	1.12	1	1	<1	0.0	0.7	23.8	0.0	21.5
EUR/USD	9.1	-0.16	1	1	1.5	0.0	0.0	0.0	19.1	4.1
GBP/USD	9.4	1.59	1	1	1.5	0.0	0.0	0.0	15.5	2.1
USD/JPY	9.8	-1.71	4	4	1.25	0.0	0.0	0.1	35.2	2.9
Bitcoin	71.0	1.35	1.25	4	<1	24.4	4.8	89.4	100.0	87.2
Coca-Cola	23.2	1.73	1.5	1.5	<1	0.0	0.2	19.8	88.9	10.0
Exxon Mobil	23.8	1.93	1.75	1.75	<1	0.0	0.0	17.8	88.1	6.8
JPMorgan	34.9	1.19	1	1	<1	0.2	0.8	60.4	99.1	42.1
Apple	43.6	1.24	1	1.25	<1	11.3	6.2	77.7	100.0	67.6
Tesla	57.2	1.48	1.5	1.5	<1	2.0	0.2	92.7	100.0	83.7
US small-cap leg (Mkt+SMB)	18.4	2.57	2.5	2.5	<1	0.0	0.1	10.9	69.3	14.4
US IG corporates (exhibit)	5.0	6.42	3.5	3.5	3.5	0.0	0.0	0.0	0.0	2.5

(Observed, C). The Dow Jones Industrial Average peaked at 381.17 on 3 September 1929 and bottomed at 41.22 on 8 July 1932, a fall of 89 percent (Observed, B); on the dividend-adjusted Shiller series the cumulative total return from October 1929 to July 1932 is −80.6 percent over 34 months (Reconstructed, A). Run through the machine at monthly frequency with era terms (10 percent maintenance, financing at an assumed 5 percent per year (Assumed, C)), the survival leverage is 2.0 times, and the manner of the survival is the finding: at 2 times the path ends at 1.98 percent of starting equity, having touched 1.77 percent, a technical survival across the 1 percent ruin line by a margin thinner than the data's own rounding, with a drawdown of about 98 percent (Modelled, B). At 2.25 times the same path ruins. The era's typical 10 times was beyond the ruin bound ($1/0.81 \approx 1.2$ times on the realised path) by an order of magnitude. The point the case adds to the framework is that at monthly frequency the binding constraint in a slow crash is not liquidation but compounding against the path: no single month had to kill the 2 times holder, the sequence nearly did.

7.2 1987: the one-day bound

The quantity is the largest leverage a single day can kill: $L = 1/|\text{worst day}|$. On 19 October 1987 the S&P 500 fell 20.47 percent (Carlson, 2007) (Observed, B), which sets the bound at 4.9 times for any holder checked intraday; the Dow fell 22.6 percent. The CRSP value-weighted series used in this paper records −17.41 percent for that day (Observed, A), and the difference matters to the machine and is reported rather than smoothed over: on the paper's own series, over the October 1987 to March 1988 window with the close-based maintenance rule, 2.5 times survives and 2.75 times is liquidated (on the

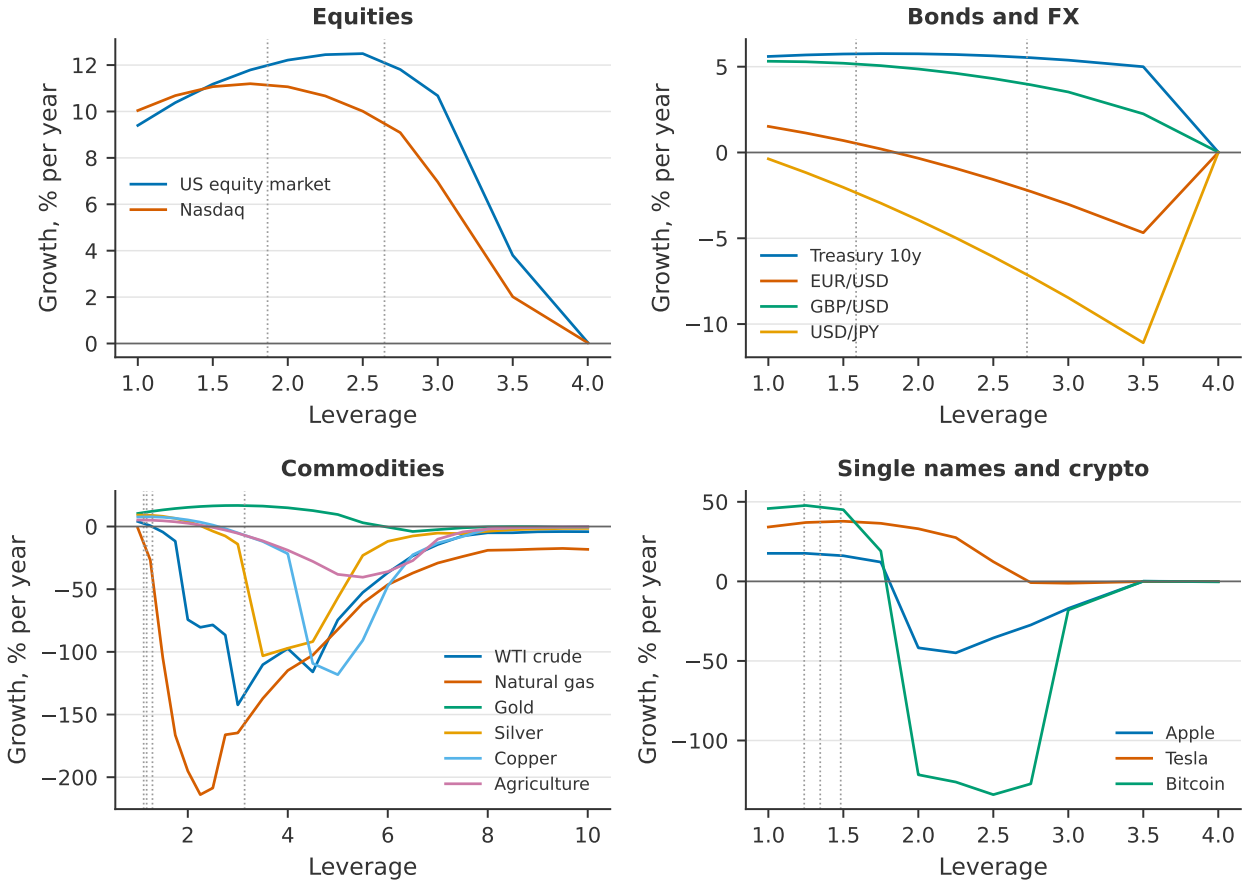


Figure 1: Annualised log growth against leverage, bootstrap engine. Dotted vertical lines mark the Gaussian Kelly fraction on the same sample. The peak of each curve is the growth-optimal leverage of Table 2. Modelled, B.

crash day itself, at its post-move equity ratio of just under the 25 percent requirement) (Modelled, A). The two numbers are the paper's lower-bound limitation in miniature: a close-based check on a −17.4 percent day passes a 2.5 times holder whom an intraday check on a −20.5 percent index would have stopped out hours earlier. Portfolio insurance, the era's mechanical seller (an estimated USD 60 to 90 billion of equity under rules that sold roughly a fifth of holdings per 10 percent fall (Observed, B)), is the same rebalancing feedback as the machine's, run in reverse: the machine buys back exposure on the way down to hold leverage constant; the insurers sold into it.

7.3 LTCM 1998: leverage set against the wrong volatility

The quantity is the multiple of book volatility the fund was effectively running, and the uniform adverse move its balance sheet could absorb. At the start of 1998 LTCM held equity of USD 4.72 billion against about USD 129 billion of assets, assets of about 27 times equity (usually quoted as about 25 times debt to equity), with derivative notionals of about USD 1.25 trillion on top (President's Working Group on Financial Markets, 1999) (Observed, B). A uniform adverse move of 3.7 percent across the book was sufficient to exhaust equity entirely (Reconstructed, B). The fund's own pre-event return volatility was a small fraction of equity-index volatility, because the book was convergence spreads; leverage had been sized, in effect, by dividing a return target by that small

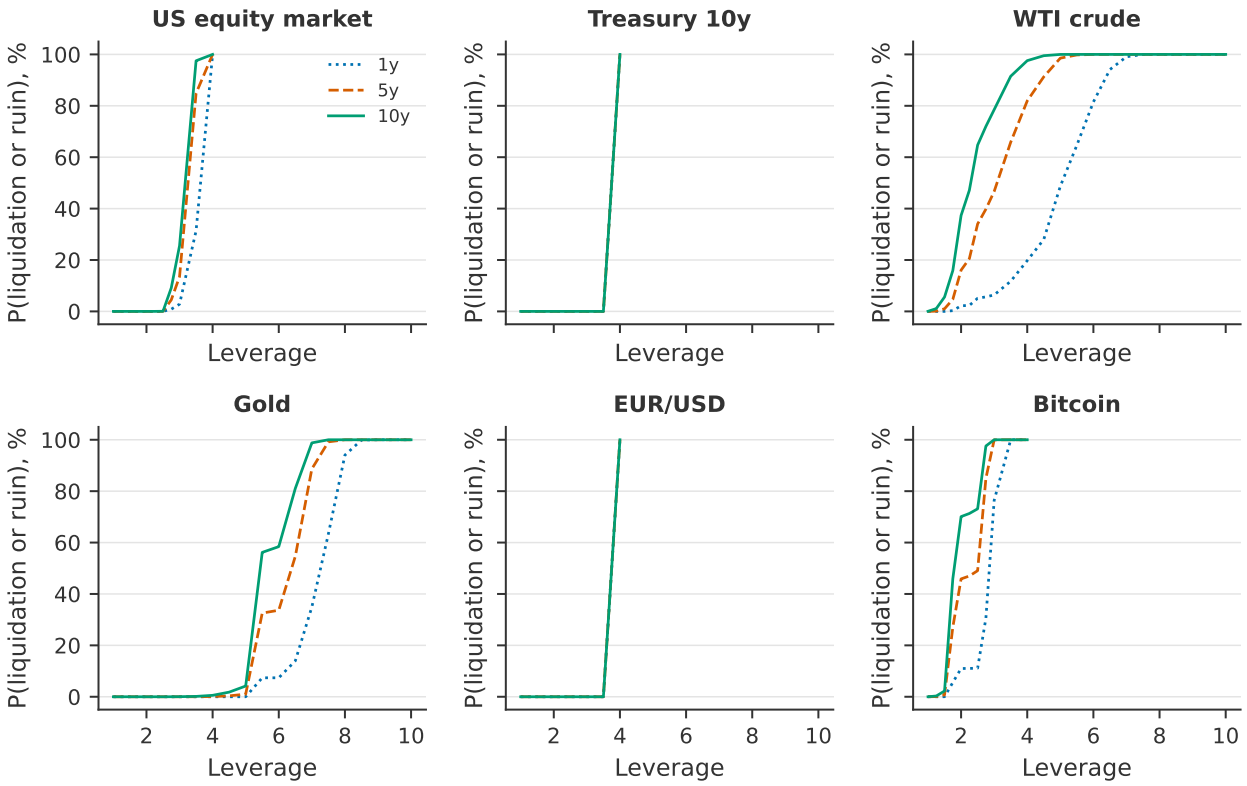


Figure 2: Probability of margin-call liquidation or ruin against leverage at 1, 5, and 10 year horizons, bootstrap engine. Above the maintenance cap ($1/m$) the liquidation engine is inadmissible and the curve is not drawn. Modelled, B.

volatility. Russia's default and moratorium of 17 August 1998 moved the spreads themselves: the fund lost USD 1.85 billion in August, about 39 percent of year-start equity in a month (Observed, B), USD 550 million of it on 21 August alone, and as equity shrank the same book was 55 times levered. The recapitalisation of 23 September 1998 (USD 3.6 billion from 14 firms for 90 percent of the fund, organised by the Federal Reserve Bank of New York (Observed, B)) arrived with equity near USD 400 million. In the framework's terms: the growth-optimal leverage formula is only as good as the σ fed into it, and LTCM is the limiting case of a σ estimated on the spread rather than on what the spread does when everyone exits together.

7.4 2008: the financing terms are a state variable

The quantity is how much survival leverage falls when financing worsens inside the event. The large US dealers ran gross leverage around 30 times at the end of 2007 (Observed, B). What failed was not first the asset side but the funding: average repo haircuts on structured debt rose from about zero in early 2007 to an average of about 27 percent across the categories in Gorton and Metrick (2012), the weakest near 45 percent (Observed, A). The machine's spiral variant encodes exactly that: once a path's drawdown from its window peak reaches 10 percent, the financing spread steps to 300 basis points over bills and the maintenance requirement rises from 25 to 35 percent for the remainder of the window. On the October 2007 to March 2009 equity window, survival leverage falls from 3.0 times under constant terms to 2.25 times under the spiral; on the reconstructed Treasury window, from 3.5 to 2.75 (Modelled, A). No plausible price path at 30 times survives either version; the case's

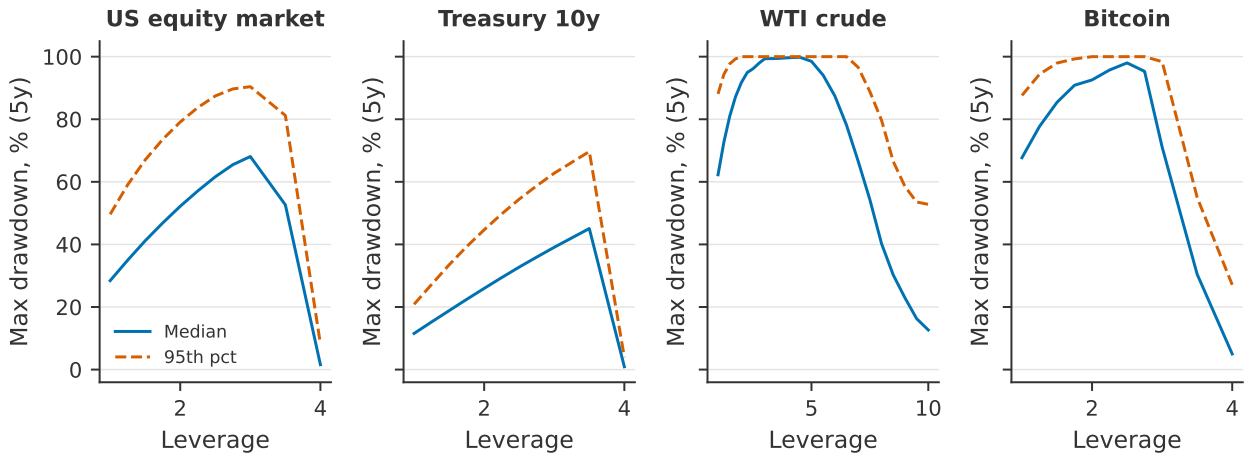


Figure 3: Median and 95th-percentile maximum drawdown within five years against leverage, bootstrap engine. Modelled, B.

contribution is the size of the financing effect at survivable leverage, roughly a quarter to a third of the survival leverage, which is the part of 2008 that a constant-terms backtest structurally cannot see.

7.5 Archegos 2021: negotiated margin and invisible aggregate leverage

The quantity is the survival leverage of concentrated single names, and the case is the panel's modern proof. Archegos, a family office exempt from the disclosure regime applied to funds, built swap exposure that the SEC's complaint puts at about USD 160 billion against capital of about USD 36 billion by March 2021, gross exposure of about 4.4 times peak capital, concentrated in a handful of names across six prime brokers, none of whom could see the aggregate (US Securities and Exchange Commission, 2022) (Observed, B). The financing term that moved first was margin generosity in the boom: Credit Suisse had cut its swap margin requirement for Archegos from 15 to 25 percent down to 7.5 percent in 2019 (Observed, B). When ViacomCBS announced a secondary offering on 22 March 2021 and fell, the calls came from all brokers at once; from 26 March the brokers liquidated into each other, ViacomCBS fell 27 percent that day, and bank losses totalled about USD 10.3 billion (Credit Suisse USD 5.5 billion, Nomura USD 2.85 billion, Morgan Stanley USD 911 million, UBS USD 774 million, MUFG USD 300 million) (Observed, B). The machine's panel supplies the denominator: the worst 252-trading-day window of each of the five panel names carries a survival leverage between 1.5 times (Apple, window from September 2000, cumulative −72.6 percent) and 3.0 times (Coca-Cola, window from October 1973), with a panel median of 2.75 times (Modelled, A). Archegos ran roughly one and a half times the worst-window survival leverage of a typical large single name, in several names at once, on margin terms that had been halved in the good years. Concentration did the rest: the names were correlated through the holder himself, since his own liquidation was the common shock.

7.6 UK LDI 2022: hedged is not the same as funded

The quantity is the collateral arithmetic of a hedged book. After the fiscal statement of 23 September 2022, the 30-year gilt yield rose about 130 basis points in three trading days, a move the Bank of England described as three times larger than any comparable move in its records, from about 3.6

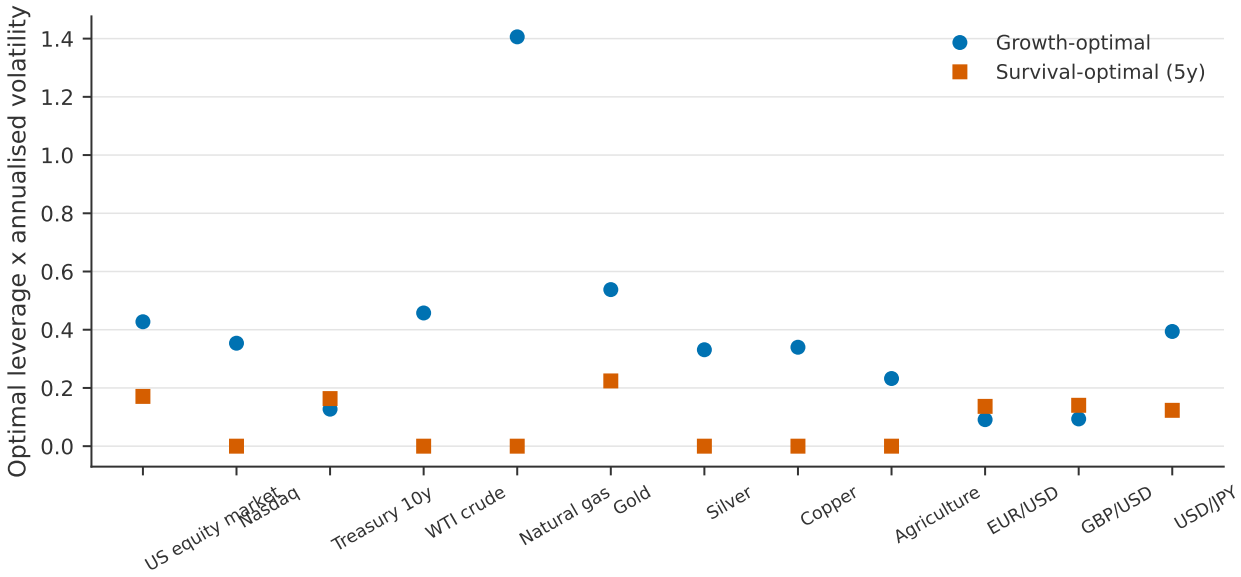


Figure 4: The unifying test: optimal leverage multiplied by annualised volatility, per core asset. A flat line would mean leverage advice reduces to volatility advice. Modelled, B.

percent to above 5 percent (Bank of England, 2022) (Observed, A). A 30-year par gilt at a 3.6 percent yield has a modified duration of 18.2, so the yield move is a price move of about −23.6 percent on the hedge asset (Reconstructed, A). Schemes' solvency improved as yields rose (their liabilities fell in value by at least as much); their collateral did not care. Variation margin and repo calls on the hedge book demanded cash of the order of a quarter of hedge notional within days, against buffers sized for moves well under 100 basis points. The Bank of England's temporary purchase facility (announced up to GBP 65 billion on 28 September; GBP 19.3 billion actually bought (Observed, A)) stopped the forced-sale loop. On the machine's terms the episode is a liquidation event at trivial price leverage: the exposure that failed was not a multiple of the schemes' assets but of their *liquid* assets, and the relevant survival constraint was the buffer, not the balance sheet. The Treasury reconstruction's own September to October 2022 window carries a survival leverage of 3.5 times for a funded holder (Modelled, A), which is precisely the contrast: the same yield path is survivable by a holder whose constraint is solvency and fatal to one whose constraint is next week's collateral call.

7.7 Nickel 2022: the short side and the limits of daily data

The quantity is the ruin bound for a short. Tsingshan Holding Group, the world's largest nickel producer, had built a short position estimated at 100,000 to 200,000 tonnes across about ten brokers (Observed, C). After Russia's invasion of Ukraine squeezed deliverable supply, LME nickel rose about 250 percent in a little over 24 hours, printing above USD 100,000 per tonne on 8 March 2022 (Oliver Wyman, 2023) (Observed, B). A short loses L times the price rise: at a 250 percent rise, every short leverage above 0.4 times is beyond total loss before any offsetting physical position (Reconstructed, B), and Tsingshan's paper loss at suspension was reported near USD 8 billion (Observed, C). The exchange then did what the machine cannot: it suspended trading on 8 March and cancelled that day's trades, converting an arithmetic ruin into a negotiated settlement. The case is also the paper's data limit made concrete. The machine runs on daily closes; the ruin happened inside two sessions and substantially intraday, and the neighbouring metals in this paper's daily data

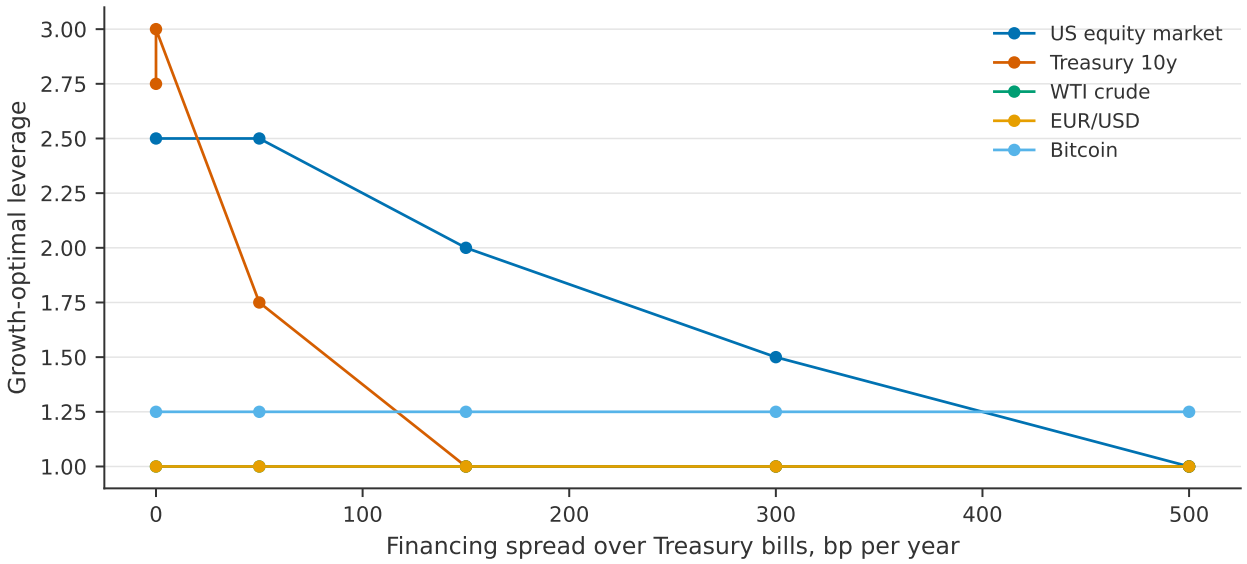


Figure 5: Growth-optimal leverage against the financing spread over Treasury bills. Points at 150 and 500 basis points use 5,000 paths; 0, base, and 300 use the main runs. Modelled, B.

show nothing like it (largest daily moves in the same fortnight: silver +4.8 percent, copper +3.3 percent (Modelled, A)). A daily machine prices the risk of the days it can see.

7.8 Volmageddon 2018: convex underlyings break the grid

The quantity is why some exposures are excluded from the grid rather than ranked by it. On 5 February 2018 the VIX rose from 17.31 to 37.32, up 115.6 percent, its largest one-day percentage rise (Observed, B). The VelocityShares XIV note, about USD 2.5 billion, was long −1 times a short-term VIX futures index that itself rose about 97 percent that day; the note lost about 96 percent of its value and was terminated under its acceleration clause (an 80 percent intraday decline trigger), and short-volatility products in aggregate shrank from about USD 3.7 billion to about USD 525 million (Observed, B). No leverage choice inside a long-only daily grid describes this position: its underlying is a futures curve on a volatility index, convex in exactly the states that matter, and its daily-reset rebalancing had to buy the rising futures into the close, feeding the spike, as the post-event analyses by Cboe and the CFA Institute document (Observed, B). The case stands in the paper as the boundary marker of the method: when the asset's own distribution is manufactured by the leveraged products trading it, the historical σ is not a property of the asset, and the Kelly arithmetic that consumes it is uninformative at exactly the wrong moment.

8 Limitations and robustness

What would prove the conclusions wrong. The paper's central claim is falsifiable in a specific way. If the pre-committed RQ4 test had found $L^*\sigma$ constant across classes (coefficient of variation at most 0.30, max/min under 2) for both optima, the correct headline would have been that there is no asset-class answer, only a volatility answer, and class-by-class leverage advice is a category error. The verdicts are reported in Section 6 for each optimum separately, with the statistic, so a reader who prefers a different tolerance can re-judge the same numbers. A second falsifier runs through the raw

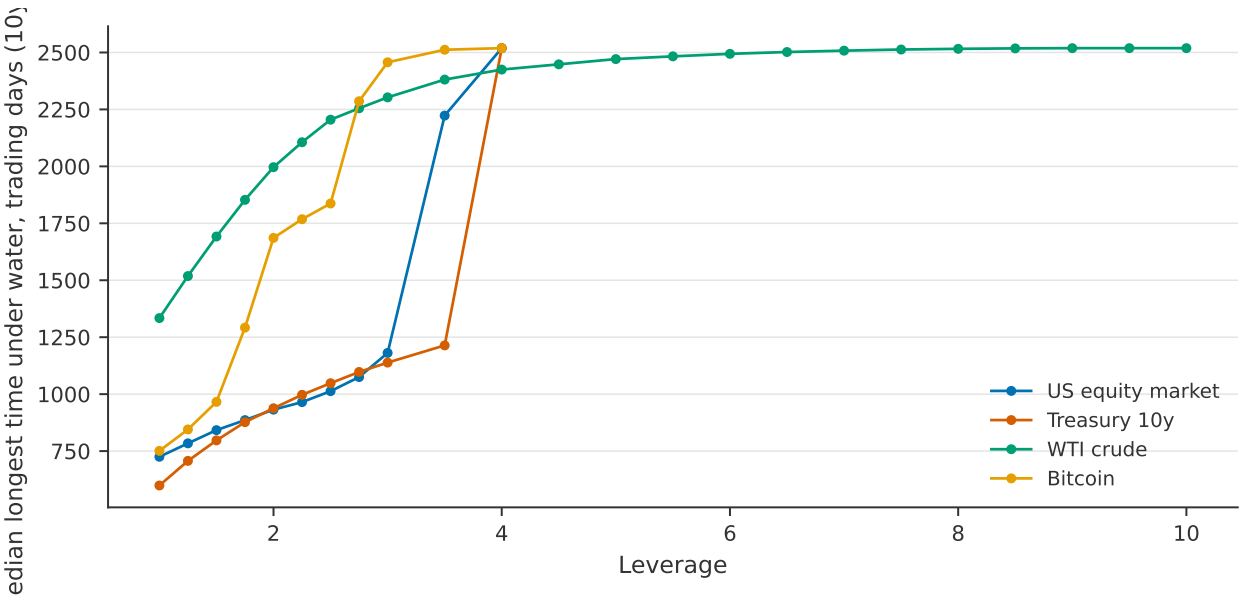


Figure 6: Median longest time under water within 10-year bootstrap paths, against leverage. Modelled, B.

engine: if the raw historical windows had produced optima far from the bootstrap's in a systematic direction (rather than noisier around them), the resampled paths, not history, would be driving the results, and the paper would have had to say so.

Close-based margin makes every risk number a lower bound. Margin is checked at closes. Intraday, positions die that the close-based rule saves: on 19 October 1987 the US market's intraday path was materially worse than its close-to-close return in the series used here, and modern brokers and clearing houses check margin continuously and can liquidate intraday. Every ruin and liquidation probability in this paper therefore understates its real-world counterpart, and the gap is largest exactly for the high-leverage, gap-prone cases. Relatedly, liquidation is assumed to execute at the breaching close; in 1987, 2008, and the 2022 nickel market, execution at the close price was not available at size, and in the nickel case not available at all once the exchange suspended trading.

Constant leverage is a contract, not a behaviour. The machine holds leverage constant, which describes a leveraged fund and a disciplined rebalancing rule. It does not describe a margin borrower who deleverages under pressure, a dealer whose balance sheet expands in calm times (Adrian and Shin, 2010), or a hedger whose notional is set by a physical exposure. The cases in Section 7 are chosen partly to mark where the constant-leverage assumption is the wrong model of the holder (LDI buffers, Tsingshan's hedge), and the results should not be read onto those holders without the translation the cases supply.

Financing is a rate here; in the world it is also a relationship. The machine's lender always lends at bill plus a spread. Real financing is withdrawn, repriced, and re-margined, which the 2008 spiral variant approximates crudely (a step change after a 10 percent drawdown) and nothing else in the paper captures. Survival leverage under live financing terms will be lower than the base case wherever the lender's terms are procyclical, which is most places.

Survivorship in the single-name panel. The five names are 2026's large survivors with long histories. A randomly chosen large 1980 stock's distribution includes delisting and failure, which this panel excludes by construction; the panel's optima and ruin probabilities are therefore optimistic for the single-name class as such, and the panel median should be read as a bound, not an estimate. The design's intended cross-check, a high-volatility French leg (market plus SMB), is reported in the

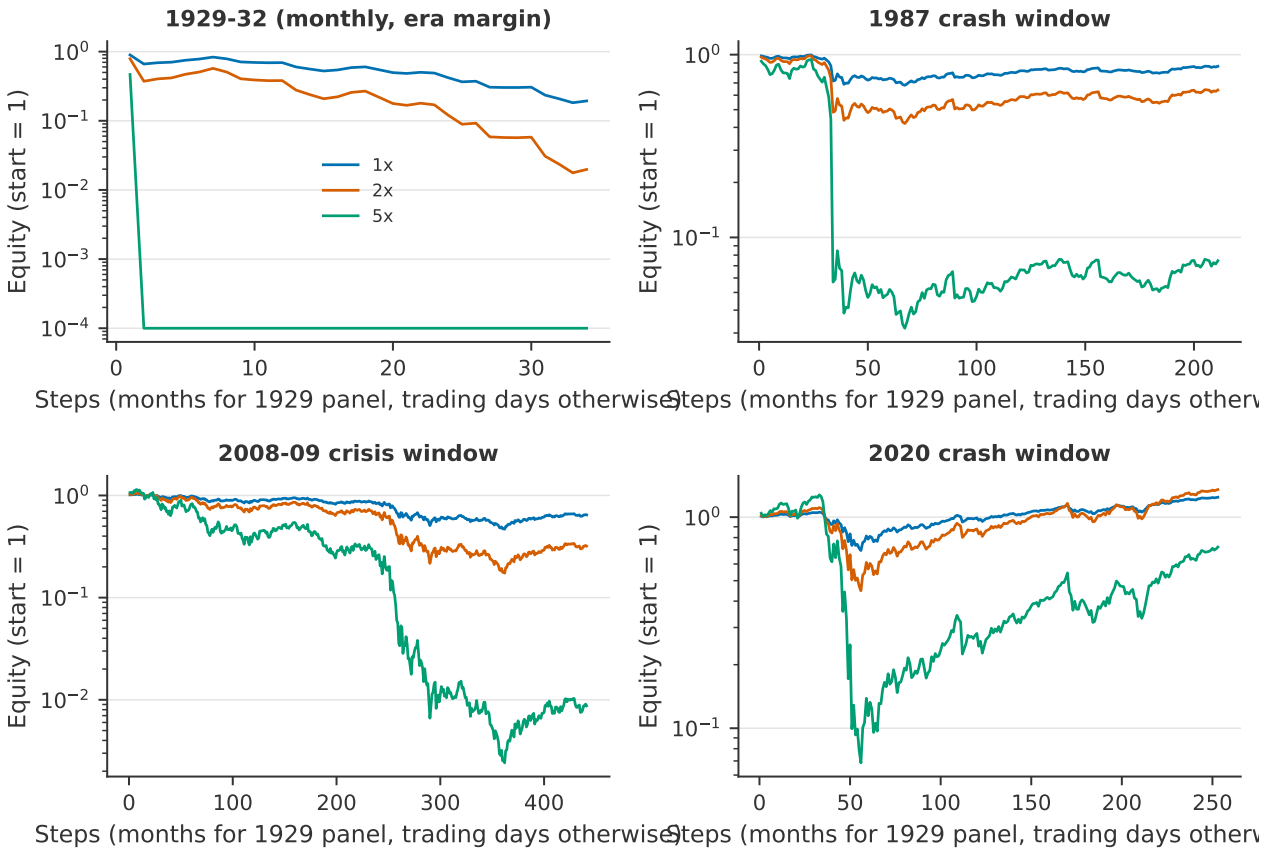


Figure 7: Equity paths at 1, 2, and 5 times in the four worst windows, log scale. The 1929 to 1932 panel is monthly (Shiller total return) at era terms (10 percent maintenance); the others are the French daily series under the base rule, under which 5 times is inadmissible and its line is the no-liquidation variant. Modelled, B.

robustness table for the same reason.

Sample and construction caveats. The futures samples begin in 2000, so the commodity optima are estimated on two and a half decades that include a commodity supercycle and its unwinding but no 1970s. Front-month futures returns exclude roll yield and financing of the futures position itself beyond the standard spread. The Nasdaq series is a price index (no dividends), biasing its growth figures down. The Treasury series is a repricing reconstruction, validated annually (Section 4) but not identical to any traded index at daily frequency. The corporate exhibit's 783 days span a single rate regime and its optima are shown for completeness under an exhibit label, never in a headline. Bitcoin is quarantined from cross-asset claims (eleven calendar years, one asset, still alive). The 1929 case runs at monthly frequency because daily data of the era is not free; monthly rebalancing understates path risk relative to the daily machine, so its survival leverage is an upper bound on the era's daily-checked equivalent.

Retrospective design. This is a retrospective analysis of historical data. The research design, including the RQ4 verdict rule and the RQ2 tolerances, was fixed by the desk before the simulations were run and is recorded in the build notes; it was not pre-registered in any public registry, and the case windows are the famous ones, chosen because they are the events the framework must explain, which is itself a selection the reader should note.

Robustness summary. Table 4 collects the sensitivity of the two optima to the liquidation rule

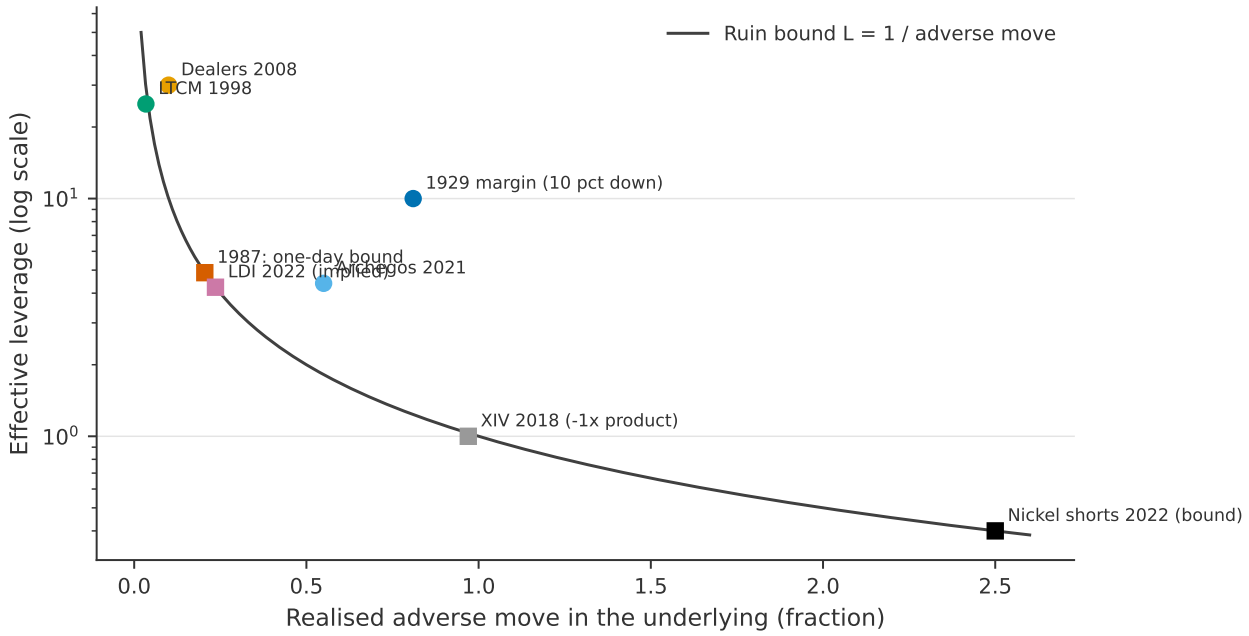


Figure 8: The eight cases against the ruin bound $L = 1/\text{adverse move}$. Points above the line hold leverage that the realised move alone was sufficient to exhaust; points on the line are bounds or implied values, as labelled, not observed pairs. Observed and Reconstructed, B.

(off: the leveraged-fund contract), to block length (5 and 63 days against the base 21), and to the financing spread (0 and 300 basis points against the class base). The results section reads the table asset by asset; the summary here is that the growth optimum is the more robust of the two (it moves mainly with the spread, as it should, since the spread is a direct tax on growth), while the survival optimum is a cliff-edge statistic by construction: it is the last grid point before a probability crosses a tolerance, so small changes in the path distribution move it by a full grid step, and it should be consumed as a range (the optimum plus and minus one step) rather than a point.

9 Conclusion

One machine, applied unchanged to every class, produces a short answer and a long one. The short answer is Table 2: growth-optimal leverage is a small multiple everywhere (equities near 2.5 times, gold the highest headline row at 3 times, the Treasury reconstruction at 1.75 times, currencies 1 to 4 times with the yen's 4 times a liquidation artefact rather than a holdable optimum, the single-name panel's median name 1.5 times), survival-optimal leverage is lower still (1 times for equities, 2.25 times for the Treasury, and below 1 time for thirteen of the twenty rows, every single name among them), and above 4 times a margined equity position cannot legally exist under the maintenance covenant at all. Anyone holding a round number far from their row of that table is not disagreeing with this paper's taste; they are disagreeing with the worst days in the sample, which is a respectable position only if taken on purpose.

The long answer is the RQ4 verdict, negative for both optima. Volatility orders the pattern: raw optima rise as volatility falls across the table. But the product of optimal leverage and volatility is not the constant a volatility rule requires, for the growth optimum (coefficient of variation 0.87, max/min 15.4) or for the survival optimum (coefficient of variation 1.09, the product at zero in seven of twelve

core classes). What volatility does not capture is the rest of the machine: the tails (which pull the simulated optimum below the Gaussian formula in fifteen of the twenty rows), the maintenance covenant (which caps holdable leverage before the return distribution gets a say), and the financing terms (which worsen inside the events that matter, at a cost of roughly a quarter of the survival leverage in the 2008 window). A volatility-only rule is therefore a good first approximation and a bad final answer, and the cases show the failure modes in the order they usually arrive: the financing moves first (2008, Archegos), then the margin call converts a paper drawdown into a realised one (1929, LDI), and the price move itself finishes only the holders the first two steps left standing (LTCM being the exception that died of the move, at 27 times, on a 3.7 percent uniform adverse move).

Two practical corollaries follow for a holder setting a multiple in advance. Keep the survival promises, not the growth peak, as the sizing constraint if the stake cannot be replaced: the growth optimum tolerates drawdowns and underwater periods (Figure 6) that most mandates, trustees, and households will not sit through, and the distance between the two optima is the price of sleeping through the path. And state the multiple against the volatility and the worst day, not against the recent past: every case in Section 7 is a holder whose leverage was reasonable against the distribution they were looking at, and ruinous against the one that arrived.

Production note. This paper was produced by Kilambi Research's quantitative research pipeline with AI assistance; all analysis was verified against source data. The replication files (data pulls with served ranges, the simulation code with its hand-check log, per-asset results, and the numerical-consistency sweep) are preserved with the working paper's build notes.

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Table 3: The eight cases on a standardised basis. Machine survival leverage is the highest grid leverage at which the machine survives the stated asset and window under the stated terms. Sources: case files assembled for this paper from the public record; grades per row in the text.

Case	Effective leverage	Vol. before event	Adverse move realised	Time to distress	Financing / margin change	Machine survival leverage
1929 margin collapse	10	21 (monthly basis)	−81% total return, Oct 1929 to Jul 1932	calls within days; trough 34 months	Fed discount rate to 6% (Aug 1929); call rates spiked	2.0 (monthly, era terms)
1987 Black Monday	2 (Reg T initial)	13	−20.5% S&P 500 in one day (−17.4% in the CRSP series used here)	1 day	none during the event	2.5 (window to Mar 1988); one-day bound 4.9
LTCM 1998	27 (assets/equity)	fund returns: low single digits monthly	book: −3.7% uniform-equivalent; Aug 1998 equity loss −39%	about 5 weeks (17 Aug to 23 Sep)	collateral demands rose as spreads widened	n/a (spread book; equity exhausted by a 3.7% uniform move)
Dealers and repo 2008	about 30	11	equity −57% peak to trough; structured marks far less liquid	weeks (acute: Sep 2008)	repo haircuts 0 to about 27% average; repo spreads 6 to 248bp	3.0 constant terms; 2.25 under the financing spiral
Archegos 2021	4.4 aggregate (to about 13 at the loosest margin)	53 (panel median, 2020)	ViacomCBS −55% peak to trough in days; −27% on 26 Mar	4 days (22 to 26 Mar)	margin had been cut to 7.5% in 2019; calls from all brokers at once	panel median 2.75 (worst 252d windows)
UK LDI 2022	hedge notional vs small liquid buffers	6 (Treasury recon., 2021)	30y gilt yield +130bp in 3 days; price about −24%	3 to 5 days	variation margin and repo calls of about a quarter of hedge notional	3.5 (funded holder, same yield path)
LME nickel 2022	short; position 100 to 200kt	metals complex, elevated	+250% in about 24 hours (short loses)	2 sessions	margin calls in the billions; exchange suspended trading and cancelled trades	0.4 (short ruin bound at a +250% move)
Volmageddon 2018	−1 (inverse product)	VIX at 17.3 before the event	VIX +116%; futures index about +97% in one day	1 day (hours)	no margin; product acceleration clause (80% intraday fall)	n/a (excluded from the grid; convex underlying)

Table 4: Robustness of the optima. Growth-optimal and 5-year survival-optimal leverage under the base case, with the liquidation rule removed, at block lengths 5 and 63 days, and at financing spreads of 0 and 300 basis points. Modelled, B.

Asset	Growth-optimal leverage			Survival-optimal (5y)		
	Base	No liq.	Blocks 5/63	Base	Spread 0	Spread 300bp
US equity market (French)	2.5	2.5	2.5/2.5	1	1	1
Nasdaq Composite	1.75	1.75	1.75/1.75	<1	<1	<1
US Treasury 10y (recon.)	1.75	1.75	1.75/1.75	2.25	2.25	2
Crude oil (WTI)	1	1	1/1	<1	<1	<1
Natural gas	1	1	1/1	<1	<1	<1
Gold (futures)	3	3	3/3	1.25	1.25	1
Silver	1	1	1/1	<1	<1	<1
Copper	1.25	1.25	1.25/1.25	<1	<1	<1
Agriculture basket	1	1	1/1	<1	<1	<1
EUR/USD	1	1	1/1	1.5	1.5	1.25
GBP/USD	1	1	1/1	1.5	1.5	1.5
USD/JPY	4	1	4/4	1.25	1.25	1.25
Bitcoin	1.25	1.25	1.25/1.25	<1	<1	<1
Coca-Cola	1.5	1.5	1.5/1.5	<1	<1	<1
Exxon Mobil	1.75	1.75	1.75/1.75	<1	<1	<1
JPMorgan	1	1	1/1	<1	<1	<1
Apple	1	1	1.25/1.25	<1	<1	<1
Tesla	1.5	1.5	1.5/1.5	<1	<1	<1
US small-cap leg (Mkt+SMB)	2.5	2.5	2.25/2.25	<1	<1	<1
US IG corporates (exhibit)	3.5	4.5	3.5/3.5	3.5	3.5	3

A Data appendix

Table 5 lists every series used, its producer, its identifier or construction, its observation count, and its sample. The financing leg for all series is the 3-month Treasury bill rate (FRED: DTB3, served from 4 January 1954), with the Kenneth R. French daily risk-free rate annualised filling earlier dates where DTB3 is unavailable; the class spread (50 basis points, 100 for single names) is added in the simulation, not in the stored series. Pull date for all series: 26 October 2026. Substitutions against the original plan (Yahoo Finance adjusted closes in place of Stooq for single names and futures; corporates demoted to a post-2023 exhibit; gold futures in place of the discontinued spot series) are recorded in Section 4 and in the replication manifest.

B Implementation checks and regime detail

B.1 Checks the implementation was required to pass

Before any production run, the simulation code was validated on hand-computable cases. At leverage 1 with no spread and no liquidation, the machine reproduces the closed form exactly (a constant 0.1 percent daily return over 2520 days gives terminal equity 12.413 against $1.001^{2520} = 12.413$). A single -20.5 percent day (the S&P 500's 19 October 1987 return) ruins a 5 times path (equity multiplier $1 - 5 \times 0.205 < 0$, terminal equity 0) and liquidates a 4 times path at the 25 percent maintenance requirement on that day with residual equity 0.18, both matching hand calculation. A 2 times path on the same day ends at 0.59, neither ruined nor liquidated. The bootstrap index generator was checked for range and block structure, and the growth ordering on synthetic data (optimum between the no-leverage and double-Kelly points) behaves as theory requires. The Treasury reconstruction was validated against published annual total returns (Section 4). The full-sample ruin accounting was also audited after a defect was found in an early version of the raw-window annualisation (a liquidated path's frozen equity annualised over the days to liquidation only, which briefly made an early-liquidated 4 times bitcoin path the spurious growth argmax); the correction (annualisation over the full sample, the liquidated path thereafter in cash) is in the code and the corrected stage was recomputed for every asset.

B.2 Regime detail (RQ3)

Table 6 reports the two optima within named subsamples and within terciles of trailing 63-day realised volatility, for the core assets, from 5,000-path bootstraps of each subsample's own days. Two patterns are visible in most rows and are stated here because the main text reads the table only in summary. In the 2008 to 2009 subsample the two optima split: growth optima rise for most core assets, because the resampled pool contains the 2009 rebound, while survival optima fall, to below 1 time for every core asset except the Treasury reconstruction (1.75 times) and EUR/USD (1 time). In the high-volatility tercile, both optima are lower than in the low-volatility tercile for nearly every asset. The 2020 subsample, a crash followed by a policy-driven recovery inside a single year, produces growth optima close to or above the full-sample ones for equities, which is a reminder that a one-year pool resampled to ten-year paths is a statement about that year's day distribution, not about holding through a decade like it.

Table 5: Series, producers, identifiers, samples. Observed, A (producer records), except the Treasury reconstruction and the agriculture basket, which are Reconstructed, A.

Series	Producer	Identifier / construction	Obs.	Sample
french_mkt	Kenneth R. French Data Library (Dartmouth)	Daily Fama/French factors, Mkt-RF + RF	26,317	1926–2026
french_smb_leg	Kenneth R. French Data Library (Dartmouth)	Daily Fama/French factors, Mkt-RF + SMB + RF	26,317	1926–2026
nasdaq	FRED (Federal Reserve Bank of St. Louis)	NASDAQCOM	14,047	1971–2026
ko	Yahoo Finance chart API	KO adjusted close	14,325	1970–2026
xom	Yahoo Finance chart API	XOM adjusted close	14,325	1970–2026
jpm	Yahoo Finance chart API	JPM adjusted close	11,747	1980–2026
aapl	Yahoo Finance chart API	AAPL adjusted close	11,559	1980–2026
tsla	Yahoo Finance chart API	TSLA adjusted close	4,106	2010–2026
wti	FRED	DCOILWTICO	9,525	1986–2026
natgas	FRED	DHHNGSP	5,711	1997–2026
gold_gc	Yahoo Finance chart API	GC=F front-month futures	6,563	2000–2026
silver	Yahoo Finance chart API	SI=F front-month futures	6,564	2000–2026
copper	Yahoo Finance chart API	HG=F front-month futures	6,567	2000–2026
corn	Yahoo Finance chart API	ZC=F front-month futures	6,574	2000–2026
wheat	Yahoo Finance chart API	ZW=F front-month futures	6,586	2000–2026
soy	Yahoo Finance chart API	ZS=F front-month futures	6,566	2000–2026
ags_basket	Computed from components above	Equal-weight daily returns of corn, wheat, soy	6,515	2000–2026
treasury10	Reconstructed from FRED DGS10	Constant-maturity 10y par bond, repriced daily (duration/convexity from yield)	16,187	1962–2026
corp_exhibit	FRED	BAMLCC0A0CMTRIV (ICE BofA US Corporate Index total return)	783	2023–2026
fx_eur	FRED	DEXUSEU spot + IR3TIB01EZM156N carry	6,973	1999–2026
fx_gbp	FRED	DEXUSUK spot + IR3TIB01GBM156N carry	13,994	1971–2026
fx_jpy	FRED	DEXJPUS spot (inverted) + IR3TIB01JPM156N carry	6,159	2002–2026
btc	FRED	CBBTCUSD (Coinbase bitcoin reference rate via FRED)	4,312	2014–2026

Table 6: Growth-optimal and 5-year survival-optimal leverage by subsample and volatility tercile (stationary bootstrap of subsample days, 5,000 paths). Modelled, B.

Asset	pre2000		2000		2007		2008		2009		2010		2019		2020		2021		present		vol low		vol mid		vol high	
	G	S	G	S	G	S	G	S	G	S	G	S	G	S	G	S	G	S	G	S	G	S	G	S	G	S
US equity	2.5	1	1	<1	4	<1	3	1.75	2.25	<1	3	1.25	3.5	2.25	3	1.5	1	<1								
Nasdaq	2.75	1	4	<1	4	<1	3	1.5	2.75	<1	2.25	1	3.5	1.25	2.25	1	1.25	<1								
Treasury 10y	1	2.25	3.5	2.75	2.5	1.75	3.5	2.75	3.5	3.5	4	1.25	1	3	3.5	2.5	1.5	1.75								
WTI	1	<1	1.25	<1	8	<1	10.0	<1	9.5	<1	1	<1	10.0	<1	1	<1	1	<1								
EUR/USD	4	<1	3.5	2	1	1	4	1.25	3.5	3.5	1	1.5	1	2	1	1.5	1	1.25								
Bitcoin	–	–	–	–	–	–	1.25	<1	1.75	<1	1	<1	2.75	<1	4	<1	1.5	<1								
Apple	1	<1	1.25	<1	1	<1	2.75	<1	2.75	<1	2.25	<1	3	1	1.25	<1	1	<1								
Gold	–	–	5	1.5	2.25	<1	1.25	1	5	1.75	3	1.25	7	1.75	7	2	1	<1								
Silver	–	–	2	<1	1	<1	1	<1	2.25	<1	1	<1	2	<1	2.75	<1	1	<1								

G: growth-optimal leverage. S: survival-optimal leverage (5y).